amazon

SALES ANALYSIS

Using SQL

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INTRODUCTION

E-commerce

Amazon was founded in 1994 as an online bookstore. Over time, it expanded into one of the world's largest e-commerce platforms, offering a vast range of products such as electronics, clothing, furniture, groceries, toys, and much more. Today, customers can browse and purchase millions of items online and have them conveniently delivered to their homes or workplaces.

WHAT IS SQL

SQL(Structured Query Language)

is a Structured Query Language (SQL) is the standard programming language used to manage and manipulate relational databases. It allows users to perform operations such as retrieving (querying), inserting, updating, and deleting data stored in a database. SQL works with data organized in tables, where rows represent individual records and columns represent specific data fields. It provides an efficient and structured way to interact with and manage large amounts of data.

PROBLEM STATEMENT

You are working at Amazon, a leading global e-commerce company. The management team has assigned you the task of analyzing sales data using SQL queries. Your responsibility is to extract meaningful insights by performing various operations such as retrieving, filtering, sorting, and aggregating data from the database to support business decisions.

QUESTIONS-:

Q1. LIST ALL PRODUCTS WITH A DISCOUNTED PRICE BELOW ₹500.

Q2.FIND PRODUCTS WITH A DISCOUNT PERCENTAGE OF 50% OR MORE.

Q3.RETRIEVE ALL PRODUCTS WHERE THE NAME CONTAINS THE WORD "CABLE."

Q4.DISPLAY THE DIFFERENCE BETWEEN THE AVERAGE OF THE ACTUAL PRICE AND THE DISCOUNTED PRICE FOR EACH PRODUCT.

Q5.QUERY REVIEWS THAT MENTION "FAST CHARGING" IN THEIR CONTENT.

Q6.IDENTIFY PRODUCTS WITH A DISCOUNT PERCENTAGE BETWEEN 20% AND 40%.

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Q7.FIND PRODUCTS THAT HAVE AN ACTUAL PRICE ABOVE ₹ ARE RATED 4 STARS OR ABOVE.

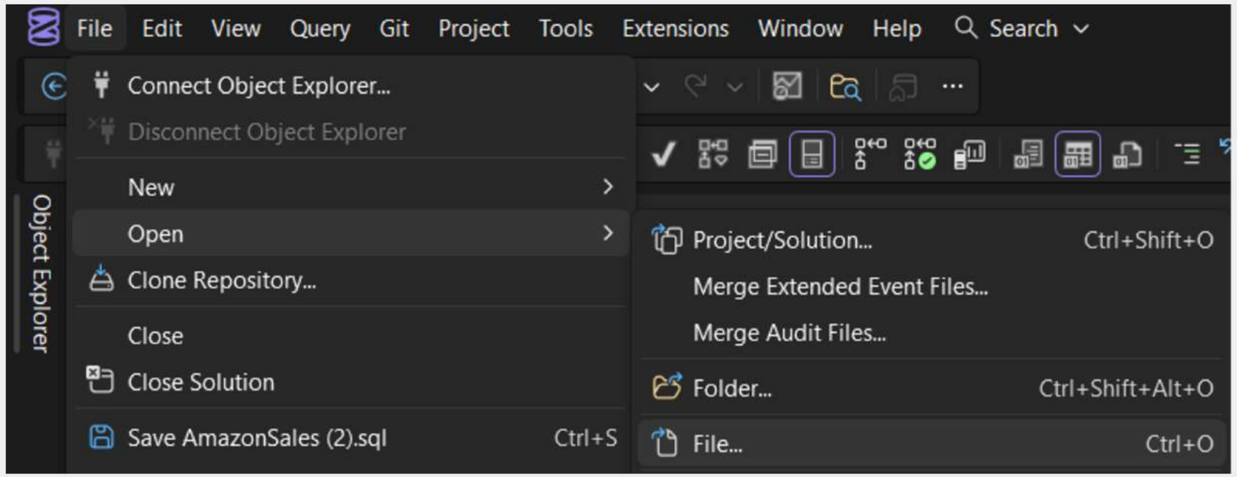
Q8.FIND PRODUCTS WHERE THE DISCOUNTED PRICE ENDS WITH A 9

Q9.DISPLAY REVIEW CONTENTS THAT CONTAINS WORDS LIKE WORST, WASTE, POOR, OR NOT GOOD.

Q10.LIST ALL PRODUCTS WHERE THE CATEGORY INCLUDES "ACCESSORIES."

GET STARTED WITH THE PROJECT

IMPORT THE DATA BY CLICKING ON FILE AND SELECTCTING OPEN THEN FILE UPLOAD YOUR DATASET FILE



AFTER THE DATASET IS UPLOADED, YOU'LL HAVE ACCESS TO THE AMAZON DATA, ALLOWING YOU TO START YOUR PROJECT

CREATE A NEW DATABASE NAMED `AMAZON` AND THEN USE IT BY EXECUTING THE COMMAND `USE AMAZON

```
create database amazon_datab

use amazon_datab
```

```
CREATE TABLE mytable (
  product_id VARCHAR(1024),
  product_name VARCHAR(1024),
  category VARCHAR(1024),
  discounted_price decimal(10,2),
  actual_price decimal(10,2),
  discount_percentage decimal(10,2),
  rating decimal(10,2) NULL,
  rating_count decimal(10,2) NULL,
  about_product Text,
  user_id VARCHAR(1024),
  user_name VARCHAR(1024),
  review_id VARCHAR(1024),
  review_title VARCHAR(1024),
  review_content Text,
  img_link VARCHAR(1024),
  product_link VARCHAR(1024)
);
```

Q1. LIST ALL PRODUCTS WITH A DISCOUNTED PRICE BELOW ₹500.

PURPOSE

This SQL query selects the product_name and discounted_price columns from a table named mytable. It filters the results to display only those records where the discounted_price is less than 500.

```
--1
select product_id,product_name,discounted_price from mytable
where discounted_price < 500;
```

product_id	product_name	discounted_price
B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Chargin...	399.00
B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...	199.00
B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync USB...	199.00
B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stre...	329.00
B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin US...	154.00
B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...	149.00
B08WRWPM...	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Ca...	176.63
B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	229.00
B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	499.00

Q2.FIND PRODUCTS WITH A DISCOUNT PERCENTAGE OF 50% OR MORE.

PURPOSE

SELECTS THE PRODUCT_ID, PRODUCT_NAME, AND DISCOUNT_PERCENTAGE COLUMNS FROM THE TABLE NAMED MYTABLE. IT FILTERS THE RESULTS TO ONLY INCLUDE ROWS WHERE THE DISCOUNT_PERCENTAGE IS GREATER THAN OR EQUAL TO 0.50.

```
--2
select product_id,product_name,discount_percentage from mytable
where discount_percentage >= 0.50;
```

	product_id	product_name	discount_percentage
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Chargin...	0.64
2	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync USB ...	0.90
3	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	0.53
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin US...	0.61
5	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...	0.85
6	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cab...	0.65
7	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	0.50
8	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3A Typ...	0.55

Q3.RETRIEVE ALL PRODUCTS WHERE THE NAME CONTAINS THE WORD "CABLE."

PURPOSE

THIS QUERY RETRIEVES SPECIFIC DATA FROM THE MYTABLE TABLE. IT SELECTS TWO COLUMNS: PRODUCT_ID AND PRODUCT_NAME. THE WHERE CLAUSE FILTERS THE RESULTS TO ONLY INCLUDE ROWS WHERE THE PRODUCT_NAME CONTAINS THE WORD "CABLE"

```sql
--3
select product_id,product_name from mytable
where product_name like '%Cable%';
```

	product_id	product_name
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Chargin...
2	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...
3	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync USB...
4	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...
5	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin US...
6	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...
7	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cab...
8	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)
9	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...
10	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3A Typ...

Q4.DISPLAY THE DIFFERENCE BETWEEN THE AVERAGE OF THE ACTUAL PRICE AND THE DISCOUNTED PRICE FOR EACH PRODUCT.

PURPOSE

THE QUERY RETURNS A LIST OF PRODUCT NAMES ALONG WITH THE DIFFERENCE BETWEEN THEIR AVERAGE ACTUAL PRICE AND AVERAGE DISCOUNTED PRICE. THIS CAN BE USED TO UNDERSTAND HOW MUCH DISCOUNT, ON AVERAGE, IS BEING APPLIED PER PRODUCT.

```
--4
select product_id,product_name,avg(actual_price)-avg(discounted_price) as differences from mytable
group by product_id,product_name;
```

	product_id	product_name	differences
1	B0BMTZ4T1D	!!1000 Watt/2000-Watt Room Heater!! Fan Heater!!Pure ...	815.000000
2	B0BMZ6SY89	!!HANEUL!!1000 Watt/2000-Watt Room Heater!! Fan Heat...	700.000000
3	B08TGG316Z	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ultra High ...	1400.000000
4	B09RFB2SJQ	10WeRun Id-116 Bluetooth Smartwatch Wireless Fitness ...	1400.000000
5	B00N1U7JXM	3M Post-it Sticky Note Cube, 200 Sheets (4 Colors x 50 S...	85.000000
6	B00N1U9AJS	3M Scotch decimal(10,2) Sided Heavy Duty Tape(1m hold...	35.000000
7	B0B9JZW1SQ	4 in 1 Handheld Electric Vegetable Cutter Set,Wireless Fo...	702.000000
8	B09F6D21BY	7SEVEN Compatible LG TV Remote Suitable for LG Non ...	1209.000000
9	B08XMSKKMM	7SEVEN® Bluetooth Voice Command Remote for Xiaomi...	1200.000000
10	B09B125CFJ	7SEVEN® Compatible for Mi tv Remote Control Original ...	450.000000

Q5.QUERY REVIEWS THAT MENTION "FAST CHARGING" IN THEIR CONTENT.

PURPOSE

THE QUERY RETURNS A LIST OF REVIEWS THAT SPECIFICALLY MENTION "FAST CHARGING". THIS CAN BE USEFUL FOR ANALYZING CUSTOMER FEEDBACK RELATED TO PRODUCTS WITH FAST-CHARGING CAPABILITIES.

```sql
--5
select product_id,product_name,review_content from mytable
where review_content like '%fast charging%';
```

Results | Messages

	product_id	product_name	review_content
1	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m ...	I ordered this cable to connect my phone to Android...
2	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stre...	Good product,long wire,Charges good,Nice,I bough...
3	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m ...	I ordered this cable to connect my phone to Android...
4	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3A Ty...	The cable is efficient in fast charging but in quick da...
5	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braided Mi...	Good product,long wire,Charges good,Nice,I bough...
6	B085DTN6R2	Portronics Konnect CL 20W POR-1067 Type-C to 8 Pi...	Definitely isn't as good as the original cord but work...
7	B083342NKJ	MI Braided USB Type-C Cable for Charging Adapter (...	I like it ????,Best charging power . I used this cable...
8	B082LSVT4B	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m ...	I ordered this cable to connect my phone to Android...
9	B08WRBG3XW	boAt Type C A325 Tangle-free, Sturdy Type C Cable ...	Check for offera before buying,18w charging cable,,...
10	B09NHVCHS9	Flix Micro Usb Cable For Smartphone (Black)	Worked on iPhone 7 and didn't work on iPhone XR,...
11	B094JNXNPV	Ambrane Unbreakable 3 in 1 Fast Charging Braided ...	The product seems and works good but it's not a fa...
12	B09W5XR9RT	Duracell USB C To Lightning Apple Certified (Mfi) Brai...	Fast charging.,Cable seems to be of good quality, n...

Q6.IDENTIFY PRODUCTS WITH A DISCOUNT PERCENTAGE BETWEEN 20% AND 40%.

PURPOSE

SELECTING DATA FROM A TABLE NAMED MYTABLE. SPECIFICALLY, IT RETRIEVES THE PRODUCT_ID, PRODUCT_NAME, AND DISCOUNT_PERCENTAGE COLUMNS FOR ROWS WHERE THE DISCOUNT_PERCENTAGE FALLS BETWEEN 0.20 (20%) AND 0.40 (40%).

```sql
--6
select product_id,product_name,discount_percentage from mytable
where discount_percentage between 0.20 and 0.40;
```

	product_id	product_name	discount_percentage
1	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	0.23
2	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	0.33
3	B082LSVT4B	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	0.38
4	B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 32LM5...	0.39
5	B08B42LWKN	OnePlus 80 cm (32 inches) Y Series HD Ready LED S...	0.25
6	B094JNXNPV	Ambrane Unbreakable 3 in 1 Fast Charging Braided Mul...	0.25
7	B0B3MMYHYW	OnePlus 126 cm (50 inches) Y Series 4K Ultra HD Sma...	0.28
8	B07KRCW6LZ	TP-Link Nano AC600 USB Wi-Fi Adapter(Archer T2U N...	0.38
9	B014I8SSD0	Amazon Basics High-Speed HDMI Cable, 6 Feet - Supp...	0.35
10	B09Q5SWVBJ	OnePlus 80 cm (32 inches) Y Series HD Ready Smart ...	0.27
11	B0B15CPR37	Samsung 108 cm (43 inches) Crystal 4K Neo Series Ult...	0.31

Q7.FIND PRODUCTS THAT HAVE AN ACTUAL PRICE ABOVE ₹1,000 AND ARE RATED 4 STARS OR ABOVE.

PURPOSE

THE SQL QUERY IN THE IMAGE IS SELECTING DATA FROM A TABLE CALLED MYTABLE. IT RETRIEVES THE PRODUCT_ID, PRODUCT_NAME, ACTUAL_PRICE, AND RATING COLUMNS FOR ROWS WHERE: THE ACTUAL_PRICE IS GREATER THAN 1000. THE RATING IS GREATER THAN OR EQUAL TO 4. THE PURPOSE OF THIS QUERY IS TO FIND PRODUCTS THAT ARE PRICED ABOVE 1000 AND HAVE A RATING OF 4 OR HIGHER.

```sql
--7
select product_id,product_name,actual_price,rating from mytable
where actual_price > 1000 and rating >= 4;
```

Results | Messages

	product_id	product_name	actual_price	rating
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Charging ...	1099.00	4.20
2	B0B6F7LX4C	MI 80 cm (32 inches) 5A Series HD Ready Smart Andro...	24999.00	4.20
3	B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 32LM5...	21990.00	4.30
4	B09C6HXFC1	Duracell USB Lightning Apple Certified (Mfi) Braided Sy...	1799.00	4.50
5	B09F6S8BT6	Samsung 80 cm (32 Inches) Wondertainment Series H...	22900.00	4.30
6	B0B1YVCJ2Y	Acer 80 cm (32 inches) I Series HD Ready Android Sm...	19990.00	4.30
7	B08B42LWKN	OnePlus 80 cm (32 inches) Y Series HD Ready LED S...	19999.00	4.20
8	B09W5XR9RT	Duracell USB C To Lightning Apple Certified (Mfi) Braid...	1999.00	4.40
9	B08QSC1XY8	Zoul USB C 60W Fast Charging 3A 6ft/2M Long Type C...	1099.00	4.30
10	B07XLCFSSN	Amazonbasics Nylon Braided Usb-C To Lightning Cable...	1900.00	4.40
11	B0B3MMYH...	OnePlus 126 cm (50 inches) Y Series 4K Ultra HD Sma...	45999.00	4.20
12	B09C6HWG18	Duracell Type C To Type C 5A (100W) Braided Sync & ...	1999.00	4.20

Q8.FIND PRODUCTS WHERE THE DISCOUNTED PRICE ENDS WITH A 9

PURPOSE

This SQL query selects product_id, product_name, and discounted_price from mytable where the discounted_price ends with 9.

The condition discounted_price % 10 = 9 ensures only those records are selected whose last digit is 9.

```
--8
select product_id,product_name,discounted_price from mytable
where discounted_price  % 10 = 9
```

	product_id	product_name	discounted_price
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Chargin...	399.00
2	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...	199.00
3	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync USB ...	199.00
4	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	329.00
5	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...	149.00
6	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	229.00
7	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	499.00
8	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...	199.00
9	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braided Mic...	299.00
10	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Black, ...	219.00
11	B09KLVMZ3B	Portronics Konnect L 1.2M POR-1401 Fast Charging 3...	159.00
12	B083342NKJ	MI Braided USB Type-C Cable for Charging Adapter (R...	349.00
13	B0B6F7LX4C	MI 80 cm (32 inches) 5A Series HD Ready Smart Andr...	13999.00

Q9.DISPLAY REVIEW CONTENTS THAT CONTAINS WORDS LIKE WORST, WASTE, POOR, OR NOT GOOD.

PURPOSE

The purpose of the query is to retrieve all rows from the mytable where the review_content contains negative words or phrases such as: “worst”,"waste","poor","not good” This helps in identifying negative reviews to analyze customer dissatisfaction.

```sql
--9
select product_id,product_name,review_content from mytable
where review_content like '%worst%'
or review_content like '%waste%'
or review_content like '%poor%'
or review_content like '%not good%';
```

Results | Messages

	product_id	product_name	review_content
1	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	Worth for money - suitable for Android auto... my ...
2	B08WRBG3XW	boAt Type C A325 Tangle-free, Sturdy Type C Cable wit...	Check for offera before buying,18w charging cabl...
3	B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 32LM5...	LG was always Good , correct delivery, good servi...
4	B085194JFL	tizum HDMI to VGA Adapter Cable 1080P for Projector,...	This connector has provided as a boon for my old ...
5	B09F6S8BT6	Samsung 80 cm (32 Inches) Wondertainment Series H...	Overall good.,TV picture ok smart betterSound ver...
6	B0B1YVCJ2Y	Acer 80 cm (32 inches) I Series HD Ready Android Sm...	About the TV - Wonderful----------...
7	B08B42LWKN	OnePlus 80 cm (32 inches) Y Series HD Ready LED S...	This OnePlus TV is great if you want a affordable ...
8	B008FWZGSG	Samsung Original Type C to C Cable - 3.28 Feet (1 Met...	Buy it,Received in good condition and is working ...
9	B0B3MMYHYW	OnePlus 126 cm (50 inches) Y Series 4K Ultra HD Sma...	I am posting this after a few weeks of using the pr...
10	B09JPC82QC	Mi 108 cm (43 inches) Full HD Android LED TV 4C | L4...	NOTE:@ If you select brand-authorized TV install...
11	B09L8DSSFH	7SEVEN® Compatible for Samsung Smart 4K Ultra HD...	Overall replacement of original samsung remote.it...
12	B07P681N66	TP-Link AC600 600 Mbps WiFi Wireless Network USB ...	I am using this on an old Mac Mini, since the Broa...

Q10.LIST ALL PRODUCTS WHERE THE CATEGORY INCLUDES "ACCESSORIES."

PURPOSE

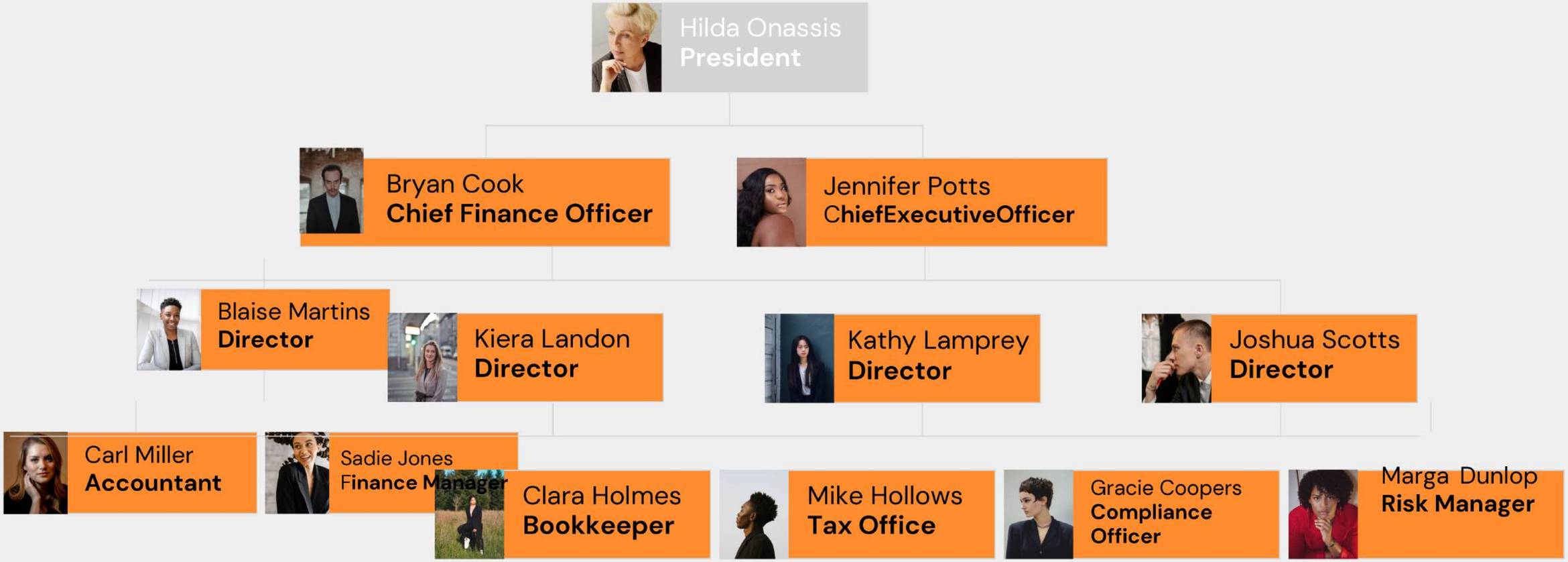
The purpose of the query is to retrieve products from mytable where the category contains the word "Accessories". This helps in filtering products related to accessories for further analysis or reporting.

```
--10
select product_id,product_name,category from mytable
where category like '%Accessories%';
```

Results | Messages

	product_id	product_name	category
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Chargin...	Computers&Accessories|Accessories&Peripherals|Cabl...
2	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...	Computers&Accessories|Accessories&Peripherals|Cabl...
3	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync USB ...	Computers&Accessories|Accessories&Peripherals|Cabl...
4	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	Computers&Accessories|Accessories&Peripherals|Cabl...
5	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin US...	Computers&Accessories|Accessories&Peripherals|Cabl...
6	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...	Computers&Accessories|Accessories&Peripherals|Cabl...
7	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cab...	Computers&Accessories|Accessories&Peripherals|Cabl...
8	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	Computers&Accessories|Accessories&Peripherals|Cabl...
9	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	Computers&Accessories|NetworkingDevices|NetworkA...
10	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m B...	Computers&Accessories|Accessories&Peripherals|Cabl...
11	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3A Typ...	Computers&Accessories|Accessories&Peripherals|Cabl...
12	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braided Mic...	Computers&Accessories|Accessories&Peripherals|Cabl...

SQL query that could summarize Amazon sales data. This query assumes you have a basic table structure for tracking products, orders, and sales.



THANK YOU

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